ITA1412 – ETHICAL HACKING

EXPERIMENT NO. 1

Experiment Name

Port Scanning Tools

Aim

To perform different types of port scans using Nmap and observe the scan output.

Commands

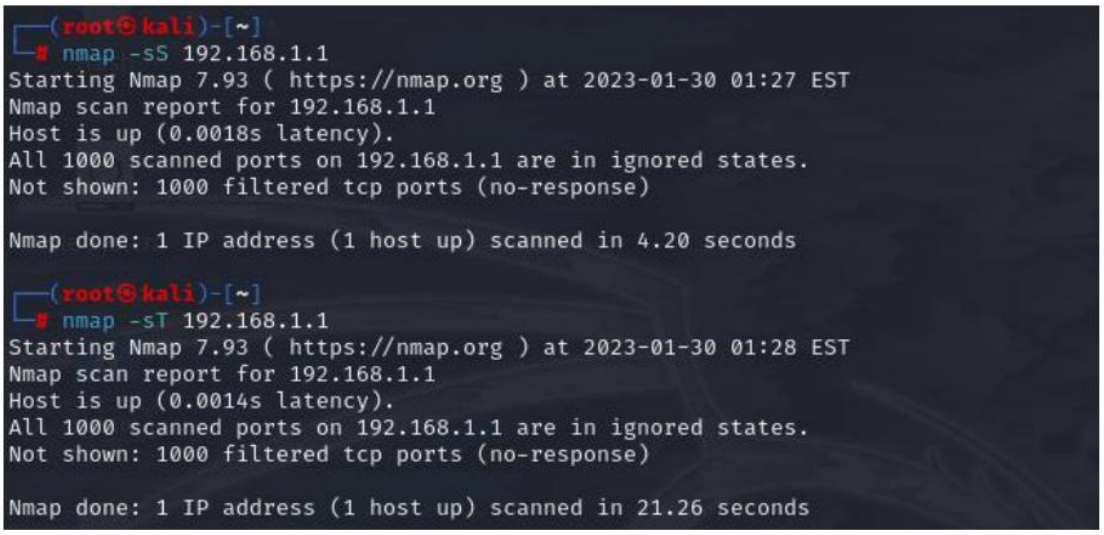
Open Nmap in Kali Linux through Applications → Information Gathering → Nmap, or run the commands in a terminal.

Scanning techniques listed in the lab manual:

Flag	Use	Example
-sS	TCP SYN port scan	nmap -sS 192.168.1.1
-sT	TCP connect port scan	nmap -sT 192.168.1.1
-sU	UDP port scan	nmap -sU 192.168.1.1
-sA	TCP ACK port scan	nmap -sA 192.168.1.1

Use these scans only on systems or networks you own or are authorized to test.

Screenshots



```
┌──(root㉿kali)-[~]
└─# nmap -sA 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 02:11 EST
Nmap scan report for 192.168.1.1
Host is up (0.00017s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds

┌──(root㉿kali)-[~]
└─# nmap -sU 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 02:12 EST
Stats: 0:04:10 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 35.90% done; ETC: 02:23 (0:07:26 remaining)
Stats: 0:04:16 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.15% done; ETC: 02:23 (0:07:32 remaining)
Stats: 0:04:30 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.80% done; ETC: 02:24 (0:07:44 remaining)
Stats: 0:04:31 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.85% done; ETC: 02:24 (0:07:44 remaining)
Stats: 0:06:01 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 40.90% done; ETC: 02:26 (0:08:42 remaining)
Stats: 0:06:06 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 41.15% done; ETC: 02:26 (0:08:43 remaining)
Stats: 0:07:10 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 44.00% done; ETC: 02:28 (0:09:07 remaining)
Stats: 0:11:52 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 30.85% done; ETC: 02:50 (0:26:36 remaining)
Stats: 0:13:26 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 32.98% done; ETC: 02:52 (0:27:18 remaining)
Stats: 0:14:39 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 34.60% done; ETC: 02:54 (0:27:41 remaining)
Stats: 0:16:08 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.58% done; ETC: 02:56 (0:27:59 remaining)
Stats: 0:16:55 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 37.62% done; ETC: 02:57 (0:28:03 remaining)
```